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# Importing required libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load the Iris dataset
iris = sns.load_dataset('iris')

# -----
# Part A: DATA ANALYSIS
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print("First 5 rows of the dataset:")
print(iris.head())

print("\nDataset Information:")
print(iris.info())

print("\nStatistical Description:")
print(iris.describe())

# Convert selected columns to NumPy array
np_array = np.array(iris[['sepal_length', 'sepal_width']])
print("\nNumPy Array (first 5 rows):")
print(np_array[:5])

# -----
# Part B: HISTOGRAMS
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plt.figure(figsize=(12, 5))

# Histogram 1
plt.subplot(1, 2, 1)
plt.hist(iris['sepal_length'], bins=15, edgecolor='black')
plt.title("Histogram of Sepal Length")
plt.xlabel("Sepal Length")
plt.ylabel("Frequency")

# Histogram 2
plt.subplot(1, 2, 2)
plt.hist(iris['petal_length'], bins=15, edgecolor='black')
plt.title("Histogram of Petal Length")
plt.xlabel("Petal Length")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()

# -----
# Part C: PAIRWISE SCATTER PLOT
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sns.pairplot(iris[['sepal_length', 'sepal_width', 'petal_length', 'petal_width', 'species']], hue='species')
plt.show()
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